HastaDAQ

Net labels connect everything: same name = same net. +3V3 / GND are plain labels, not power symbols.

Pimoroni Pico Plus 2 (+ RM2 on SP/CE): same 40-pin header as Pico 2 W. I2C0 = GP4/GP5 (header pins 6/7), to the IMU Qwiic-breakout header and to the 4-wire loom for the strain-gauge board. Crossed-out pins are intentionally unused.

J1
IMU via Qwiic breakout (The Pi Hut 106676)
GND 1 GND
+3V3 2 3V3
SDA 3 SDA
SCL 4 SCL

J2
GAUGE BOARD LOOM (J_I2C): GND, +5V, SDA, SCL
GND 1 GND
+5V 2 5V
SDA 3 SDA
SCL 4 SCL

Strain-gauge board: 4-wire I2C loom. Pin 2 is +5V, NOT 3V3 -- use a keyed plug, never a Qwiic cable. Pull-ups live on the ADS122C04 breakouts on that board.

U1
Pimoroni Pico Plus 2 (header pins)
1 GP0_UART0-TX
2 GP1_UART0-RX — SBUS_IN_GPIO
3 GND — GND
4 GP2
5 GP3
6 GP4_I2C0-SDA — SDA
7 GP5_I2C0-SCL — SCL
8 GND — GND
9 GP6
10 GP7 — ENC4_CS
11 GP8 — ENC3_CS
12 GP9 — ENC2_CS
13 GND — GND
14 GP10_SPI1-SCK — SPI_SCK
15 GP11_SPI1-TX — SPI_MOSI
16 GP12_SPI1-RX — SPI_MISO
17 GP13_SPI1-CSn — ENC1_CS
18 GND — GND
19 GP14 — TRIG_IN_GPIO
20 GP15 — TRIG_OUT_GPIO
40 VBUS
39 VSYS — +5V_VSYS
38 GND — GND
37 3V3_EN
36 3V3(OUT) — +3V3
35 ADC_VREF
34 GP28
33 AGND — GND
32 GP27
31 GP26
30 RUN
29 GP22
28 GND — GND
27 GP21_UART1-RX — SBUS
26 GP20_UART1-TX — SBUS
25 GP19
24 GP18
23 GND — GND
22 GP17
21 GP16 — SOFTSTOP_IN

J3
ENC1 SPI port: MT6835 (pitch) / Orbis
GND 1 GND
ENC1_CS 2 CS
SPI_SCK 3 SCK
SPI_MOSI 4 MOSI
SPI_MISO 5 MISO
+3V3 6 3V3
+5V 7 5V
GND 8 GND

J4
ENC2 SPI port: MT6835 / Orbis
GND 1 GND
ENC2_CS 2 CS
SPI_SCK 3 SCK
SPI_MOSI 4 MOSI
SPI_MISO 5 MISO
+3V3 6 3V3
+5V 7 5V
GND 8 GND

J5
ENC3 SPI port: MT6835 / Orbis
GND 1 GND
ENC3_CS 2 CS
SPI_SCK 3 SCK
SPI_MOSI 4 MOSI
SPI_MISO 5 MISO
+3V3 6 3V3
+5V 7 5V
GND 8 GND

J6
ENC4 SPI port: generic SPI device
GND 1 GND
ENC4_CS 2 CS
SPI_SCK 3 SCK
SPI_MOSI 4 MOSI
SPI_MISO 5 MISO
+3V3 6 3V3
+5V 7 5V
GND 8 GND

CS pull-ups (10k)
+3V3 — R1 10k — ENC1_CS
+3V3 — R2 10k — ENC2_CS
+3V3 — R3 10k — ENC3_CS
+3V3 — R4 10k — ENC4_CS

U2
Pololu D42V55F8 8.4V 5A buck (servo rail)
VSERVO 1 VOUT — 2 VOUT VSERVO
GND 3 GND — 4 GND GND
GND 5 GND — 6 GND GND
+12V 7 VIN — 8 VIN +12V
9 VRP — 10 VRP
11 PG — 12 EN

U3
SparkFun BabyBuck 5V (AP63357) (Pico VSYS, Orbis)
+12V 3 VIN — 1 VOUT_5V +5V
GND 2 GND
4 EN

D1
Schottky (e.g. 1N5822, 3A)
+5V — A K — +5V_VSYS

C1 100uF 25V input bulk (Panasonic ECA1EM101) +12V / GND
C2 470uF 16V servo bulk (Panasonic EEUFR1C471) VSERVO / GND
C3 100uF 10V 5V bulk (RS PRO LHK 5x7) +5V / GND

J7
SBUS servo header 3x6 (6 plugs: GND, V, SIG)
GND 1 S1_GND
VSERVO 2 S1_V
SBUS 3 S1_SIG
GND 4 S2_GND
VSERVO 5 S2_V
SBUS 6 S2_SIG
GND 7 S3_GND
VSERVO 8 S3_V
SBUS 9 S3_SIG
GND 10 S4_GND
VSERVO 11 S4_V
SBUS 12 S4_SIG
GND 13 S5_GND
VSERVO 14 S5_V
SBUS 15 S5_SIG
GND 16 S6_GND
VSERVO 17 S6_V
SBUS 18 S6_SIG

TEST POINTS (plain 2.0 mm pads): two GND pads for a scope clip (near the supply and near the Pico). Rails and signals are probed at component leads and header pins (C1/C2/C3, D1, Pico pins 36/39).

GND — TP1 TP
GND — TP2 TP

PWR_FLAG: marks nets supplied through the supply connector (+12V, GND) and the Schottky (+5V_VSYS)

#FLG1 PWR_FLAG — +12V
#FLG2 PWR_FLAG — GND
#FLG3 PWR_FLAG — +5V_VSYS

J10
SBUS IN (RC receiver): GND, V, SIG
GND 1 GND
+5V 2 V
SBUS_IN 3 SIG

SBUS_IN — R13 1k series — SBUS_IN_GPIO
SBUS_IN_GPIO — R14 10k pull-down (DNP at first) — GND
D4 5.1V zener (BZX55C5V1) SBUS IN clamp: GND — A K — SBUS_IN_GPIO

SBUS IN: RC-receiver pass-through (plug order GND, V, SIG). V = +5V (NOT VSERVO; the Futaba R3008SB takes 4.8–7.4 V, so 8.4 V would overstress it; check any other receiver's range). SIG -> 1k -> GP1 (UART0 RX); 5.1 V zener clamps a receiver that outputs more than 5.1 V (about 100 pF at 100 kbaud; harmless). 10k pull-down R: leave UNFITTED on the first trial; RP2350 erratum E9 can hold a pin with a weak pull-down (>~8.2k) near 2 V, so if it is fitted and misbehaves use 4.7k.

J8
Main supply cable (2x4): VSUPPLY, VGND, TRIG, SOFTSTOP
+12V 1 VSUPPLY
GND 2 VGND
TRIG_IN 3 TRIG_IN
SOFTSTOP_CABLE 4 SOFTSTOP+
GND 5 SOFTSTOP-
TRIG_OUT 6 TRIG_OUT
GND 7 VGND
+12V 8 VSUPPLY

SOFTSTOP_CABLE — R5 1k (soft-stop series) — SOFTSTOP_IN
+3V3 — R6 10k (soft-stop pull-up) — SOFTSTOP_IN
D2 5.1V zener (BZX55C5V1) soft-stop clamp: GND — A K — SOFTSTOP_IN

TRIGGER IN: any 3.3 V .. 24 V level, opto input

TRIG_IN — R7 10k pull-down — GND
TRIG_IN — R8 1k 1W — TRIG_IN_LED
U4 FOD817A opto (TRIG_IN): TRIG_IN_LED — 1 A K 4 — TRIG_IN_GPIO; GND — 2 A K 3 — GND
+3V3 — R9 10k pull-up — TRIG_IN_GPIO
D3 1N4148 (LED reverse protect): GND — A K — TRIG_IN_LED

TRIGGER OUT: open-collector NPN, pull-up rail chosen by jumper

TRIG_OUT_GPIO — R10 2.2k (base) — TRIG_OUT_BASE
TRIG_OUT_BASE — R11 10k (base-emitter) — GND
Q1 2N3904 NPN (TO-92, pin 1 E, 2 B, 3 C): TRIG_OUT_BASE — 2 B C 3 — TRIG_OUT; GND — 1 E
TRIG_PU — R12 1.5k pull-up — TRIG_OUT
J9 TRIG_OUT pull-up (1x5): +3V3 1 3V3; TRIG_PU 2 PU; +5V 3 5V; TRIG_PU 4 PU; +12V 5 12V

Shunt pins 1–2 (3V3), 2–3 (5V) or 4–5 (12V) to pull TRIG_OUT up to that rail; fit none if the receiver has its own pull-up. Pins 2 and 4 are both TRIG_PU. Fit ONE shunt only (two shunts could join rails). Pull-up current: 3.3 V ~2 mA, 5 V ~3 mA, 12 V ~7 mA (enough for a 6.5 mA opto input such as MPS4200). 24 V receivers: no jumper, add an external 24 V pull-up in the harness. 3V3 for eSync 2 high-Z ports (3.3 V max).

NOTES
1. ONE common GND (Pico, sensors, servo PSU). SBUS needs a shared ground reference.
   RM2 WiFi breakout (off-board, cabled to the Pico Plus 2 SP/CE connector): uses GP32–GP36 only, no overlap with I2C (GP4/5) or any other signal.
2. Strain gauges: SEPARATE board (kicad/gaugeboard), reached by a 4-wire I2C loom (GND, +5V, SDA, SCL; 0.1in headers, keyed plugs).
   That board makes its own clean 3.3 V from +5V and holds both ADS122C04s, the gauge headers and the filtering.
   ADS #2 needs its ADDR0 jumper bridged (expected address 0x41, unconfirmed; firmware scans at boot). I2C pull-ups come from the ADS122C04 breakouts on the gauge board and from the IMU breakout.
3. SBUS: GP20 (TX) and GP21 (RX) on one net (UART inverted in firmware). One 3x6 servo header (6 plugs of GND, V, SIG; footprint HastaDAQ:PinHeader_3x06): V = VSERVO (8.4 V). SIG = SBUS (all six in parallel). GND.
4. POWER: 12 V in, MAX 20 V (C1 is rated 25 V; silkscreen this at J8) (no fuse; the bench supply's current limit is the protection) -> +12V rail -> (a) Pololu D42V55F8 = VSERVO 8.4 V (servos), (b) SparkFun BabyBuck 5V = +5V (Orbis, gauge-board loom, Schottky -> Pico VSYS). VSYS max is 5.5 V. USB (VBUS) is left for programming: with USB only, +5V is not live.
   BabyBuck EN is left open (its own divider enables it). 5V bulk: module has 2x22uF; 100uF added near the Pico/ports.
5. Triggers: TRIG_IN accepts 3.3–24 V (opto). TRIG_OUT is open collector with a jumper-selected pull-up. Pin order: see each symbol.
   Main supply cable (2x4, pins 1–8 row-major): 1,8 VSUPPLY (12 V) | 2,7 VGND | 3 TRIG_IN | 6 TRIG_OUT | 4 SOFTSTOP+ | 5 SOFTSTOP-.
   SOFTSTOP+ -> 1k -> GP16 (10k pull-up to 3V3, 5.1 V zener to GND). SOFTSTOP- = GND: loop closed = run, broken = stop.
   The 2x4 connectors (this one and ENC1–4) are plain 0.1in headers for now: use keyed/shrouded plugs: a rotated plug would put 5V on a CS pin. Measured system current was <= 1 A at 8.4 V, well inside ~3 A per 0.1in pin.
6. Encoders: 4 x 2x4 headers (pins 1–8: GND, CS, SCK, MOSI, MISO, 3V3, 5V, GND) on one SPI1 bus; CS = GP13/GP9/GP8/GP7.
   MT6835 uses 3V3 (3.3 V logic); RLS Orbis uses 5V with 3.3 V logic. Check MISO tri-states when CS is high.
   The four ENC headers are a generic SPI port (SPI1: SCK GP10, MOSI GP11, MISO GP12, own CS each, 3V3 and 5V supply), so any 3.3 V SPI device can be plugged in, not just encoders. Each device needs its own CS and an MISO that tri-states when CS is high.

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